

Flex Sight — Temperature Monitoring & Alert System

STAGE 1 REPORT

This report documents the formation, ideation, and design of **Flex Sight** — a lightweight embedded temperature monitoring system built around a clear core workflow: **MQTT/API** → **Server** → **Dashboard Alert**. The system is designed to detect abnormal temperature levels early, trigger instant alerts at approximately **50°C**, and improve safety response across operational environments such as server rooms, warehouses, and industrial facilities.

Team Formation & Roles

The project team was structured around the main system workflow — **MQTT/API** → **Server** → **Dashboard Alert** — with each role aligned to a specific component of the pipeline. Team members introduced their technical backgrounds, strengths, and interests in embedded systems, backend development, IoT communication, dashboard development, and technical documentation.

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Project Lead & System Architecture

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Embedded Systems & Sensor Integration

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Server & Backend Development

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Dashboard & Alert Management

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Documentation & Presentation

Rabea Younis Thabit

Collaboration Strategy

Communication Tools

- **WhatsApp** — quick updates and daily communication
- **Discord** — technical discussions and file sharing
- **Google Meet** — remote meetings and team alignment
- **In-person meetings** — brainstorming and progress reviews

Documentation & Decision-Making

- **Google Docs** — collaborative report writing and editing
- **Canva** — presentation slides and visual materials
- **GitHub** — file organisation and version control

Decisions are made through group discussion. If consensus is not reached, the team uses **voting** to select the most suitable option based on feasibility, clarity, and alignment with project goals.

Brainstorming & Idea Evaluation

The team conducted structured brainstorming sessions focused on real-world problems in temperature monitoring, fire risk detection, and early safety alerts. Several techniques were used to explore and compare ideas.



Mind Mapping

Organised main concepts, system components, communication methods, and alert features visually.



"How Might We" Questions

Explored practical ways to detect abnormal temperature levels and send alerts efficiently.

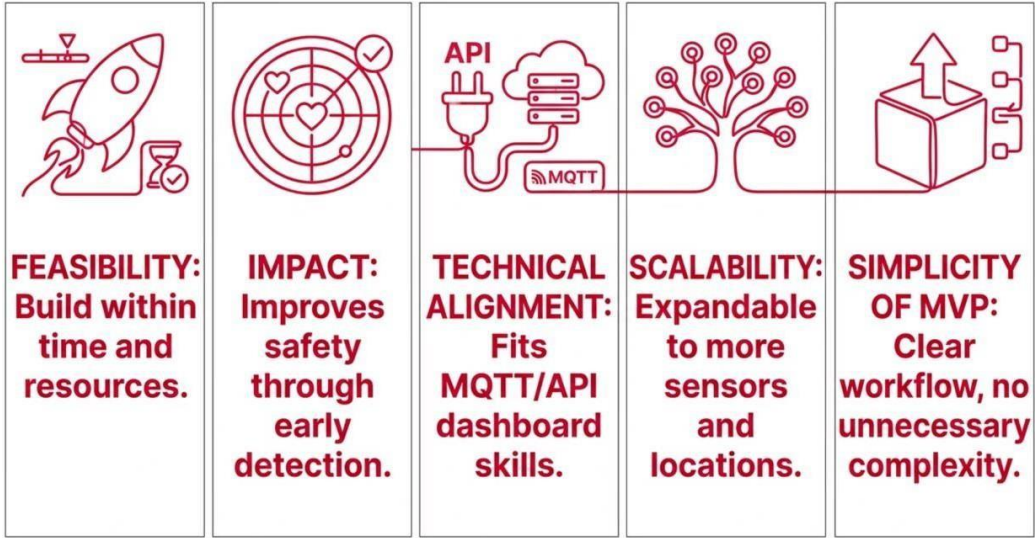


Research-Based Discussions

Focused on safety challenges, delayed fire detection, and the importance of real-time dashboard alerts.

SCAMPER

Applied the SCAMPER technique to explore how existing monitoring systems could be Substituted, Combined, Adapted, Modified, Put to other uses, Eliminated, or Reversed to better fit the Flex Sight requirements and improve the core workflow.



Ideas Explored & Rejected

Idea	Description	Strengths	Weaknesses
Smart AI Camera	Camera-based system to detect risks using AI analysis, identifying visual heat signatures and environmental hazards in real time.	High impact potential; capable of detecting multiple risk types visually; scalable to many locations.	Requires advanced AI model development; high implementation cost; complex hardware setup beyond MVP scope.
Full Fire Detection System	System combining heat, smoke, and flame detection using multiple sensors to provide comprehensive fire risk monitoring across facilities.	Comprehensive safety coverage; high detection accuracy; addresses multiple risk types simultaneously.	Requires multiple sensor types; more hardware testing needed; exceeds current timeline and team capacity.
Mobile Alert Application	Mobile app that sends alerts directly to users' smartphones, enabling remote monitoring and instant notifications outside the main dashboard.	Improves accessibility; enables remote monitoring; familiar technology for users.	Adds complexity without delivering core monitoring; dashboard alerts already cover the essential use case for MVP.
Flex Sight	Temperature monitoring system using MQTT/API for real-time data transmission, server-side processing and threshold checking, and an instant dashboard alert at approximately 50°C.	Simple to implement; technically aligned with team skills; low cost; scalable MQTT/API architecture; directly solves the safety problem.	Depends on stable internet; requires sensor calibration; fixed 50°C threshold may need adjustment per environment.

Evaluation Rubric / Scoring

Idea	Feasibility	Impact	Technical Alignment	Scalability	MVP Simplicity	Total
Smart AI Camera	2	5	3	5	2	17
Full Fire Detection System	3	5	4	5	2	19
Mobile Alert Application	4	3	4	3	4	18
Flex Sight	5	4	5	5	5	24

Risks & Constraints

Ideas with higher scores also carry higher risks. Advanced concepts such as Smart AI Camera and Full Fire Detection System exceed the current MVP timeline and team capacity. The Mobile Alert Application adds complexity without delivering the core monitoring functionality. Flex Sight balances feasibility with impact, making it the lowest-risk, highest-value choice for the first MVP stage.

Reasons for Rejection

Idea	Reason for Rejection
Smart AI Camera	Too complex for the current MVP stage and requires advanced AI model development.
Full Fire Detection System	Requires multiple sensors and more hardware testing than the timeline allows.
Mobile Alert Application	Useful, but not essential for the first MVP because dashboard alerts are enough

Final Decision:

Based on the evaluation results, the team selected **Flex Sight** as the final MVP because it is feasible, simple to implement, technically aligned with the team’s skills, and scalable for future development. The idea focuses on a clear safety problem and provides a practical workflow: **MQTT/API → Server → Dashboard Alert**.

Project Overview: Flex Sight

Flex Sight is a simple embedded monitoring system designed to continuously measure temperature levels and support early safety alerts in operational environments. Temperature data is transmitted through **MQTT/API** to the **server**, where readings are processed and checked against a defined threshold. When the temperature reaches approximately **50°C**, the system automatically triggers an instant **dashboard alert**.

Expected Outcomes

- Continuous real-time temperature monitoring across operational environments
- Automatic dashboard alert triggered when temperature reaches 50°C
- Centralised visibility of device status and historical temperature readings
- Scalable foundation for future expansion with additional sensors and AI integration

Lightweight Monitoring

Continuous, real-time temperature data collection from embedded sensors.

Centralised Processing

Server receives, processes, and evaluates all incoming temperature readings.

Instant Alerts

Dashboard alerts triggered automatically when the 50°C threshold is reached.

Problem Statement & Proposed Solution

The Problem

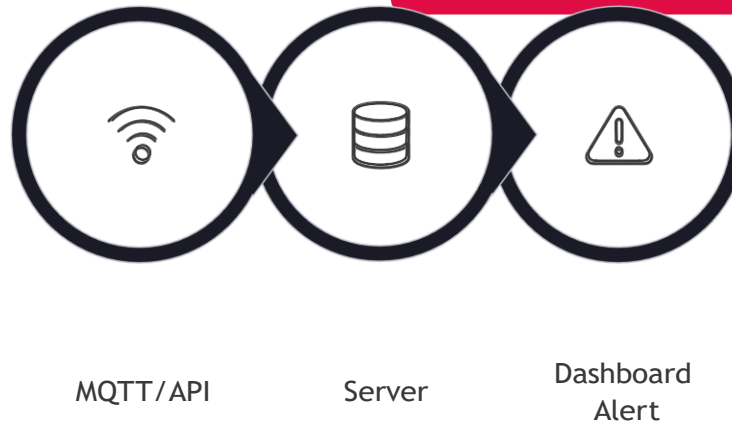
Many operational environments — server rooms, warehouses, factories, and offices — face safety risks from high temperatures and delayed monitoring. Traditional manual inspection methods mean:

- Temperature levels are not monitored continuously
- Critical heat thresholds may go unnoticed
- Alerts are delayed, unavailable, or not centralised
- Monitoring data is not easily processed through one dashboard

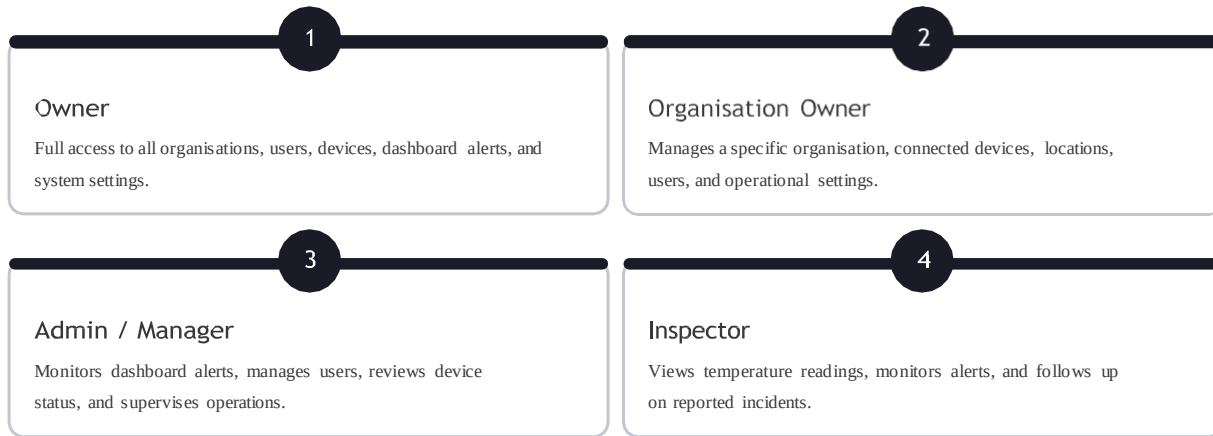
The Solution

A temperature monitoring system that follows the workflow
MQTT/API → Server → Dashboard Alert:

- Transmits data through **MQTT/API**
- Processes readings on the **server**
- Checks against the defined threshold
- Sends an automatic alert at approximately **50°C**
- Displays live readings and alerts on the **dashboard**



User Roles & Target Sectors



The platform targets sectors where high temperatures pose operational risk, including **warehouses, factories, server rooms, data centres, technical facilities, offices, industrial environments, and electrical rooms.**



Technical Stack



MQTT/API

The communication layer for lightweight, fast, real-time data transmission from the monitoring device to the server. Supports continuous temperature readings and alert triggers.



Server

Receives data from MQTT/API, processes readings, checks the **50°C** threshold, and manages alert logic, user access, and device information.



Database System

Connected to the server to store temperature readings, alerts, device records, and system logs. Supports future reporting and incident tracking.



Dashboard Alert

The user-facing monitoring interface displaying live readings, device status, and instant alerts when abnormal temperatures are detected.

Strengths, Weaknesses & Added Value



Strengths

- Simple and achievable MVP
- Real-time monitoring and dashboard alerts
- Low-cost implementation
- Scalable MQTT/API architecture
- Suitable for multiple industries
- Supports future sensor, camera, and AI expansion



Weaknesses

- Depends on stable internet connectivity
- Requires sensor calibration and testing
- Accuracy depends on hardware quality
- Fixed **50°C** threshold may need environment-specific adjustment
- Requires continuous power supply



For Organisations

Faster response, improved safety, reduced equipment damage risk, and centralised dashboard visibility.



For Safety Teams

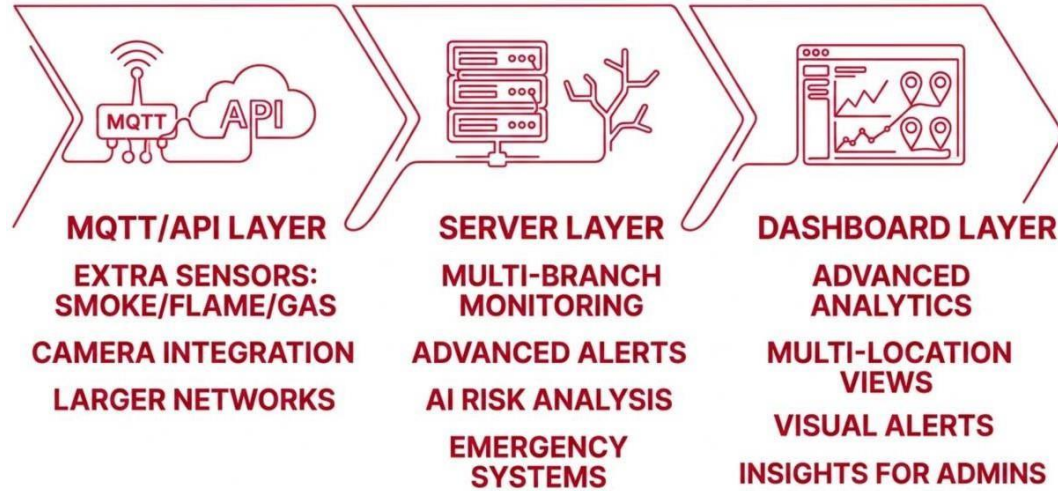
Immediate alerts, easier incident monitoring, and faster decision-making from real-time data.



For Future Development

Expandable to smoke, flame, and humidity sensors; AI integration; multi-location monitoring; and camera support.

Future Expansion & Decision Justification



The team selected Flex Sight because it addresses a **clear operational safety need**, can be built using simple and available components, and fits the project timeline and team capabilities. The idea provides a focused MVP based on real-time temperature monitoring and dashboard alerts, uses **MQTT/API** for fast and lightweight data transmission, and can be expanded in the future to support additional sensors, multiple locations, and advanced alert features. Overall, the solution is **simple, scalable, and directly connected to a real safety challenge**.

③ Core workflow: **MQTT/API** → **Server** → **Dashboard Alert** — transmit, process, and display temperature data with an automatic alert at approximately 50°C.